**Master's Thesis**

# Korea University Master Thesis & Ph.D. Dissertation Template Sample

**First Last**

**Department of Artificial Intelligence**

**Graduate School**

**Korea University**

**February 2027**

# Korea University Master Thesis & Ph.D. Dissertation Template Sample

by
First Last

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under the supervision of
Professor Advisor Name

A thesis submitted in partial fulfillment of the
requirements for the degree of
Master of Science

**Department of Artificial Intelligence**

**Graduate School**

**Korea University**

**October 2026**

The thesis of First Last has been approved by the thesis committee in partial fulfillment of the requirements for the degree of Master of Science

November 2026

_____

Committee Chair: First Professor Name

_____

Committee Member: Second Professor Name

_____

Committee Member: Third Professor Name

# Korea University Master Thesis & Ph.D. Dissertation Template Sample

by First Last
Department of Artificial Intelligence
under the supervision of Professor Advisor Name

## Abstract

English abstract goes here

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

**Keywords**: keyword one, keyword two, keyword three

# 국문초록

## Korea University Master Thesis & Ph.D. Dissertation Template Sample

First Last
Artificial Intelligence
지도교수: Advisor Name

여기에 국문 초록 내용이 들어갑니다.

**주제어**: keyword one, keyword two, keyword three

To my family and friends.

# Preface

Preface text goes here

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

# Acknowledgement

Acknowledgement goes here

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

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# Nomenclature

$c$ Speed of light in vacuum

**LSTM** Long Short-Term Memory

**GPU** Graphics Processing Unit

# Chapter 1. How to Use This Template

## 1.1 Introduction

This document explains how to use the `KUThesis` LaTeX template for Korea University thesis/dissertation submissions, following the official 2022 Korea University graduate school guidelines. This lab's copy has been adapted from the original `KUThesis2022` template by Suhyeon Lee (`https://github.com/hyeonleee/KUThesis2022`) and is kept as an English-only, internal starting point: fill in your own details and replace the placeholder chapters with your real content.

## 1.2 Key Features

The template lets you switch between a master's thesis or a Ph.D. dissertation, double-sided or single-sided printing, and a **library submission format**, so you can avoid repeating formatting work during writing, examination, and submission.

## 1.3 Compiling

This class has been tested with `pdflatex` (Overleaf and local TeX distributions such as TeX Live/MacTeX both work).

## 1.4 Other Ways to Compile

Any TeX Live 2013 or later distribution should work. On Unix-like systems, run `make`; on Windows, run `make.bat`. Use `make clean` or `clean.bat` to remove the generated build files. Keeping backups of your work is strongly recommended.

## 1.5 Project Files

This template includes the following files:

- Class file: `KUThesis.cls`
- Main file: `thesis.tex`
- English abstract: `abstract/abstract.tex`
- Korean abstract: `abstract/abstract-kr.tex` (required by regulation even for an English-language thesis - see `krabst` below)
- Acknowledgement (사사): `acknowledgement.tex`
- Dedication page (감사의 글, optional): `dedication.tex` - enable with `\includeDedication`
- Preface (서문, if necessary): `preface.tex` - enable with `\includePreface`
- Nomenclature (기호설명, optional): `nomenclature.tex` - enable with `\includeNomenclature`
- Bibliography database: `library.bib`

## 1.6 Class Options

You can pass the following options to `\documentclass[...]{KUThesis}`:

### 1.6.1 Available Options

- `doctor` / `master`: Ph.D. dissertation or Master's thesis
- `final` / (omit for draft): final submission version or a work-in-progress draft
- `library`: formats the document for digital library submission
- `twosides` / `oneside`: double-sided or single-sided printing
- `krabst`: generates the Korean abstract page (required by regulation, even for an English-language thesis)
- `science` (default) / `arts`: (master's only) labels the degree "Master of Science" or "Master of Arts"
- `asym`: adjusts the binding-side margin for print shops that prefer an asymmetric gutter
- `chapterinline` (default) / `chapterstacked`: "Chapter 1. Title" on one line, or "Chapter 1" / "Title" on two lines

### 1.6.2 Examples

- `[doctor, final, twosides, krabst, library]`: Ph.D. dissertation, library submission, double-sided, with Korean abstract (recommended)
- `[master, final, twosides, krabst]`: Master's thesis, final bound copy, double-sided, with Korean abstract (recommended)
- `[master, final, oneside]`: Master's thesis, final version, single-sided
- `[doctor, oneside]`: Ph.D. dissertation, single-sided, draft

## 1.7 Chapter Heading Style

Chapters in this template (this one included) are started with a plain `\chapter{Title}` (automatic numbering). How the heading itself looks is controlled by a class option:

- `chapterinline` (default): "Chapter 1. Title" on a single line
- `chapterstacked`: "Chapter 1" and "Title" on two separate lines (plain LaTeX report style)

Pick whichever style you prefer in `\documentclass[...]{KUThesis}`; it applies to every `\chapter` in the document automatically, so you never need to format individual chapter headings by hand.

## 1.8 What You Need to Edit

To use this template for your own thesis/dissertation, update the following in `thesis.tex`:

- `\title`, `\author`, `\advisor`: title, your name, and your advisor's name
- `\graduateSchool` / `\department`: your department and graduate school
- `\graduateDate`, `\submitDate`, `\approvalDate`: graduation, submission, and examination-approval dates
- `\referee`: your committee members' names
- `\keywords`: the keywords shown at the bottom of the abstract
- `abstract/abstract.tex` and `abstract/abstract-kr.tex`: your English and Korean abstracts
- `dedication.tex`, `preface.tex`, `nomenclature.tex` (all optional): only needed if you enable the matching `\include...` command
- Replace the placeholder chapters (this one included) under `chapters/` with your actual content

If the automatic line breaks in `KUThesis.cls` look awkward for a particularly long name, title, or department, you may need to manually adjust line breaks (`\\`) or font sizes in the relevant macro.

## 1.9 Credits

- Adapted from KUThesis2022 by Suhyeon Lee (https://github.com/hyeonleee/KUThesis2022)
- Further references:
  - KU Nuclear Physics Lab's earlier LaTeX template (https://github.com/KUNPL/KUThesis)
  - Korea University Graduate School thesis format notice: https://graduate.korea.ac.kr/academic/dissertation.html

## 1.10 Disclaimer

This template is provided as a convenience, with no warranty of any kind. The maintainers are not responsible for any issues, rejected submissions, or formatting problems that result from using it. Requirements can change and may vary by advisor, library staff, department, or program.

**Always check the latest official guidelines with your own department/graduate school before relying on this template, and leave yourself enough time before your submission deadline.**

# Chapter 2. Introduction

The following formatting information is intended to illustrate several acceptable ways of preparing a thesis or dissertation for your convenience.

Chapter 2 is styled with `\chapter{Introduction}`.

You can put `\label{chap:intro}` to refer to this chapter.

## 2.1 Second Level Heading

Section 2.1 is styled with `\section{Second Level Heading}`.

You can put `\label{sec:section}` and `\ref{sec:section}` to label and refer to this section. Sections will appear in the Table of Contents, automatically.

### 2.1.1 Third Level Heading

Subsection 2.1.1 is styled with `\subsection{Third Level Heading}`. Subsections will appear in the Table of Contents, automatically.

For more information about headings, refer to `https://www.overleaf.com/learn/latex/Headers_and_footers`

This template isn't the only way to list titles, subheadings, numbering, etc. It's just one example that may work for you and it is not mandatory or even recommended.

## 2.2 Referencing Headings

Suppose that you want to refer to the first section. The first section (of the first chapter) was labeled with `\label{sec:section}`. You can refer to the section by typing `\ref{sec:section}` : Section 2.1

Suppose that you want to refer to the first subsection. The first subsection (of the first section of the first chapter) was labeled with `\label{subs:subsection}`. You can refer to the subsection by typing `\ref{subs:subsection}` : Subsection 2.1.1

For more information about labeling and referencing, refer to the followings.

- `https://en.wikibooks.org/wiki/LaTeX/Labels_and_Cross-referencing`
- `https://www.overleaf.com/learn/latex/Cross_referencing_sections%2C_equations_and_floats`

# Chapter 3. Format

## 3.1 Paper Size and Margins

The paper sizee of the thesis/dissertation shall be B5. For the first three preliminary pages (including the cover page, title page, and signature page) before the abstract, all margins (top, bottom, left, and right) shall be at least 3 cm. From the abstract on, the top and bottom margins shall be at least 3cm, and the left and right margins shall be at least 2 cm (Table 3.1).

The paper size and margins are governed by the geometry package. For more information, refer to the following

- `http://mirrors.ctan.org/macros/latex/contrib/geometry/geometry.pdf`
- `https://www.overleaf.com/learn/latex/Page_size_and_margins`

Table 3.1: Organizing and formatting thesis/dissertation

| **Order** | **Note** | **Margin** | **Pagination** |
|---|---|---|---|
| Cover page | | top, bottom, left & right at least 3 cm | None |
| Title page | | | |
| Signature page | | | |
| Abstract | both English & Korean | top & bottom at least 3cm<br>left & right at least 2 cm | i, ii, iii, iv, $\cdots$ |
| Dedication page | optional | | |
| Preface | if necessary | | |
| Acknowledgements | optional | | |
| Table of contents | | | |
| List of tables | if there are tables or figures in the main body | | |
| List of figures | | | |
| Nomenclature | optional | | |
| Main body | | | 1, 2, 3, 4, $\cdots$ |
| Reference | | | |
| Appendices | optional | | |
| Index | optional | | |

Table 3.2: Requirement for font size and the style used in the LaTeXtemplate

| | Size requirements | LaTeX command | LaTeX size |
|---|---|---|---|
| Thesis title | 21 | `\huge` | 20.74 |
| The school name (Graduate School, Korea University) | 18 | `\LARGE` | 17.28 |
| All other parts are 16 points (department, name, advisor, master's thesis, ···, submitted, ··· completed, etc.) | 16 | `\fontsize {16pt}{16pt} \selectfont` | 16 |
| Year, month and day | 14 | `\Large` | 14.4 |
| Main Text | 10–12 | `\normalsize` | 10.95 |
| Heading | None | | |
| Figure caption | None | | |
| Table caption | None | | |

## 3.2 Fonts and Size

The default font size is set to 11pt. In LaTeX, you can use commands like `\normalsize`, `\Large`, `\LARGE`, `\huge`, and so on, to specify the size of the font. We relate the above commands to 11pt, 14pt, 18pt and 21pt, respectively, of the MS word template. In addition, the following command was used to express 16pt.

```
\fontsize{size}{baselineskip}\selectfont
```

Thus, there are slight differences in font size in MS word template and in LaTeX template (Table 3.2).

## 3.3 Tables and Figures

The font, size, alignment method, numbering method, etc. of table or figure titles can be modified, appropriately. For example, ¡Table 1¿ and ¡Figure 1¿ can also be used. Also, the style of the table (thickness and color of the border, etc.) can be modified. It is common to place figure titles below the figure and table titles above the table.

To make a table, use the environment `tabular` and specify the columns. The above table has three center-aligned columns ;

```
\begin{tabular}{ccc} ... \end{tabular}
```

You can also use an advanced version of `tabular`, which are `taubularx`, `tabulary`, `tabu`, `multirow` or `booktabs` to manipulate the typeset of tables.

It is desirable to put the `tabular` environment inside the `table` environment. You can add a caption of the table by `\caption{...}`. The labeling `\label{...}` for future reference should be followed just after the caption. All the tables in the `table` environment will be included in the 'List of Tables'.

For more information about tables, refer to

- `https://www.overleaf.com/learn/latex/Tables`

It is also possible to use LaTeX tables generator.

- `https://www.tablesgenerator.com/`

To include a figure file in the document, you can use `includegraphics` command, which requires `graphicx` package.

```
\includegraphics[width=.2\textwidth]{kumark.png}
```

You can specify the width or the height of the figure inside the square brackets and the file name (with or without the extension) inside the braces.

It is desirable to put the `includegraphics` command inside the `figure` environment. Again, the labeling needs to be followed just after the caption. All the tables in the `figure` environment will be included in the 'List of Figures'.




Figure 3.1: Korea University Global Symbol

For more information about figures, refer to the following

- `https://www.overleaf.com/learn/latex/Inserting_Images`
- `https://www.overleaf.com/learn/latex/How_to_Write_a_Thesis_in_LaTeX_(Part_3)%3A_Figures%2C_Subfigures_and_Tables`

## 3.4 Equations

You can type an equation with inline math mode like $E = mc^2$. Or you can type

$$E = mc^2$$

to express the equation in display math mode. The above equation is unnumbered. To number the equation automatically, you can use `equation` environment;

$$E = mc^2 \tag{3.1}$$

The number or the tag of the above equation reads 'the first equation of the chapter 3'. If you add one more equation, you can get the second equation of the chapter 3.

$$e^{i\theta} = \cos\theta + i\sin\theta. \tag{3.2}$$

You can also specify the tagging explicitly, using `\tag{...}`

$$E = mc^2 \tag{$*$}$$

To express a list of equations, you can use the `gather` environment, which just enumerates equations vertically. For example, suppose that you want to express a system of linear equations $x+y+z=3$, $x-y+2z=1$, $x+3z=2$. Using `gather` environment, you get

$$x+y+z=3 \tag{3.3}$$
$$x-y+2z=1 \tag{3.4}$$
$$x+3z=2. \tag{3.5}$$

If you want to un-number the equations, use `gather*` environment;

$$x+y+z=3$$
$$x-y+2z=1$$
$$x+3z=2.$$

Note that the above system is not well aligned. To align the equations horizontally, with respect

to the equality sign, you can use `align` (or `align*`) environment

$$\begin{aligned} x+y+z &= 3 \\ x-y+2z &= 1 \\ x+3z &= 2. \end{aligned}$$

`align` environment tags every equation of the system

$$x+y+z = 3 \tag{3.6}$$
$$x-y+2z = 1 \tag{3.7}$$
$$x+3z = 2. \tag{3.8}$$

If you want one tagging for the system, you can use the `aligned` environment and the `equation` environment, simultaneously ;

$$\begin{aligned} x+y+z &= 3 \\ x-y+2z &= 1 \\ x+3z &= 2. \end{aligned} \tag{3.9}$$

Finally, you can label and refer to an equation, by `\label{...}` and `\eqref{...}`. For example, you can say that 'The root of (3.9) is $x = 2$, $y = 1$, $z = 0$'. `gather` and `align` are the environments provided by the `amsmath` package. For more information to typeset the equation neatly, refer to `http://www.ams.org/arc/tex/amsmath/amsldoc.pdf`.

## 3.5 Footnotes and Endnotes

Footnotes[1] can be included to provide additional information about the content. Footnotes should be placed at the bottom of the page separated from the text by a solid line and referenced through a superscript number.

[1]The usage of footnotes is different or limited depending on the field of study. The usage of footnotes is recommended only when you're sure how a footnote should be used in your field.

## 3.6 Quotation

If you want to cite from the bibliography, you can type, for example, `\cite{LSTM}` where `LSTM` is the name of the reference: [**?** ]. Or you can cite the other reference here like this; [**?** ].

For direct quotation, you can use either the `quote` environment or the `quotation` environment.

> "Learn from yesterday, live for today, hope for tomorrow. The important thing is not to stop questioning."
>
> — Albert Einstein

> "Learn from yesterday, live for today, hope for tomorrow. The important thing is not to stop questioning."
>
> — Albert Einstein

## 3.7 Definitions and Theorems

If you want to make definitions and theorems in the paper, use the predefined (in the preamble) environments `definition` and `theorem` which are supported by the `amsthm` package.

You can either specify the name of the definition

**Definition 1** (Right Triangles)**.** *A right triangle is a triangle in which one angle is a right angle.*

or not (don't specify the name of the definition)

**Definition 2.** *A right triangle is a triangle in which one angle is a right angle.*

Here are examples of theorems ;

**Theorem 1** (Pythagorean theorem)**.** *Consider a right triangle where c is the length of the hypotenuse, and a and b are the lengths of the remaining two sides. Then*

$$a^2 + b^2 = c^2 \tag{3.10}$$

**Theorem 2.** *Consider a right triangle where c is the length of the hypotenuse, and a and b are the lengths of the remaining two sides. Then*

$$a^2 + b^2 = c^2 \tag{3.11}$$

For later use, we put indexings for a right traingle and the Pythagorean theorem here.

Sometimes you need to special font for mathematical use. For example, you may need symbols like $\mathbb{R}$, $\mathscr{T}$, $\mathscr{A}$ or $\mathfrak{M}$. Some symbols are typeseted without declaring any packages, while others need packages like amssymb or mathrsfs. For more information about typsetting mathematical expressions, refer to the followings ;

- `https://www.overleaf.com/learn/latex/Mathematical_expressions`
- `https://www.overleaf.com/learn/latex/Subscripts_and_superscripts`
- `https://www.overleaf.com/learn/latex/Brackets_and_Parentheses`
- `https://www.overleaf.com/learn/latex/Matrices`
- `https://www.overleaf.com/learn/latex/Integrals%2C_sums_and_limits`
- `https://www.overleaf.com/learn/latex/Display_style_in_math_mode`
- `https://www.overleaf.com/learn/latex/Mathematical_fonts`

# Chapter 4. Conclusion

The conclusion starts here.